ABBA6 Implicit

Yoshida seven-stage composition of projected ABBA2 maps

1 Seven-stage composition

Let $\mathcal{P}_{q,t}$ be one complete symmetric projected ABBA2 step with signed duration q. The sixth-order model uses

$$a = 0.78451361047755726382, \qquad b = 0.23557321335935813368,$$
$$c = -1.17767998417887100695, \qquad d = 1.31518632068391121889,$$

and $(w_1, \ldots, w_7) = (a, b, c, d, c, b, a)$. Starting from $z^{(0)} = z_n$, it applies

$$t_j = t_n + h \sum_{\ell<j} w_\ell, qquad z^{(j)} = \mathcal{P}_{w_j h, t_j}(z^{(j-1)}), \qquad z_{n+1} = z^{(7)}.$$

The binary64 coefficients satisfy, up to round-off,

$$\sum_j w_j = 1, \qquad \sum_j w_j^3 = 0, \qquad \sum_j w_j^5 = 0.$$

Together with palindromy, these are the relevant odd-order cancellation conditions for a symmetric second-order base.

2 Projection at every factor

Every $\mathcal{P}_{w_j h, t_j}$ starts a new projection problem. In reduced form, with the usual diagonal operators $E, P, G, N = G^T$, factor j solves

$$M_j(\mu_j) = \Psi_{w_j h, t_j}(E z^{(j-1)} + N\mu_j), \tag{1}$$
$$G M_j(\mu_j) + 2\mu_j = 0, \tag{2}$$
$$z^{(j)} = P(M_j(\mu_j) + N\mu_j). \tag{3}$$

Thus one outer step has seven projections and seven nonlinear solves. It is not a single projection around an unprojected seven-map base.

3 Configurations

The class supports two algebraically equivalent residual formulations (reduced and simultaneous), two nonlinear solvers (Newton and Broyden), and physical or fully extended state. Physical execution may additionally enable the triangular $\kappa = k/2$ energy sidecar; fully extended execution always enables energy evolution. The chosen values apply uniformly to all seven factors. For N physical guiding-centre particles, with tracking either off or on, a reduced solve has a $2N$-component multiplier and a $4N$-component duplicated base state. Fully extended execution currently requires one particle.

4 Signed time path

Coefficients c in factors three and five are negative. Partial sums also leave the outer interval: the composition reaches approximately $t_n + 1.02009h$, then $t_n - 0.15759h$, later $t_n + 1.15759h$, and finally $t_n + h$. This non-monotone path is part of the real sixth-order composition; the dynamics must be defined at all internal times.

5 Order and finite-root qualifications

If the projected ABBA2 base is symmetric and second order, and each projection equation is solved to the relevant local root, the palindromic composition has designed global order six. Finite Newton or Broyden stopping, floating-point coefficient error, and non-smooth dynamics perturb this ideal statement. Diagnostics consequently retain work and residual data for every factor as well as outer-step aggregates.

6 Implementation correspondence

The public class is in `src/simulation/methods/abba/order6_implicit.py`; coefficients are in `src/simulation/methods/_abba_coefficients.py`; and the general signed-composition coordinator is shared with `ABBA4Implicit`. The companion architecture document under `../simulation/` gives the complete runtime and observation contract.